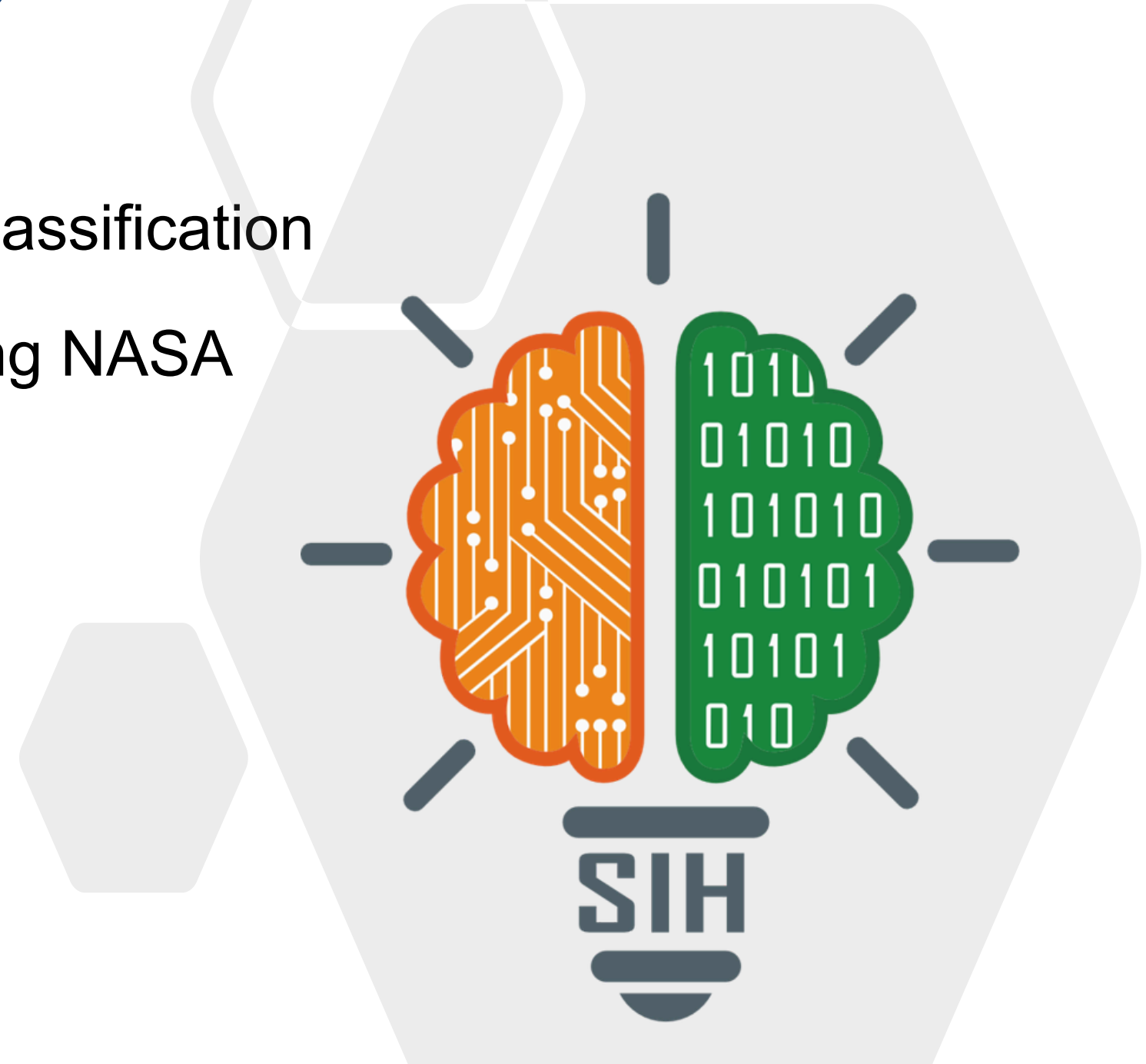


THERMALINTELL - AI-Powered Geospatial Intelligence for Industrial Fire & Thermal Anomaly Detection

- **Problem Statement ID** - SIH26162
- **Problem Statement Title** - AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data
- **Theme** - Disaster Management
- **PS Category** - Software
- **Team ID** -
- **Team Name** - Deadlock

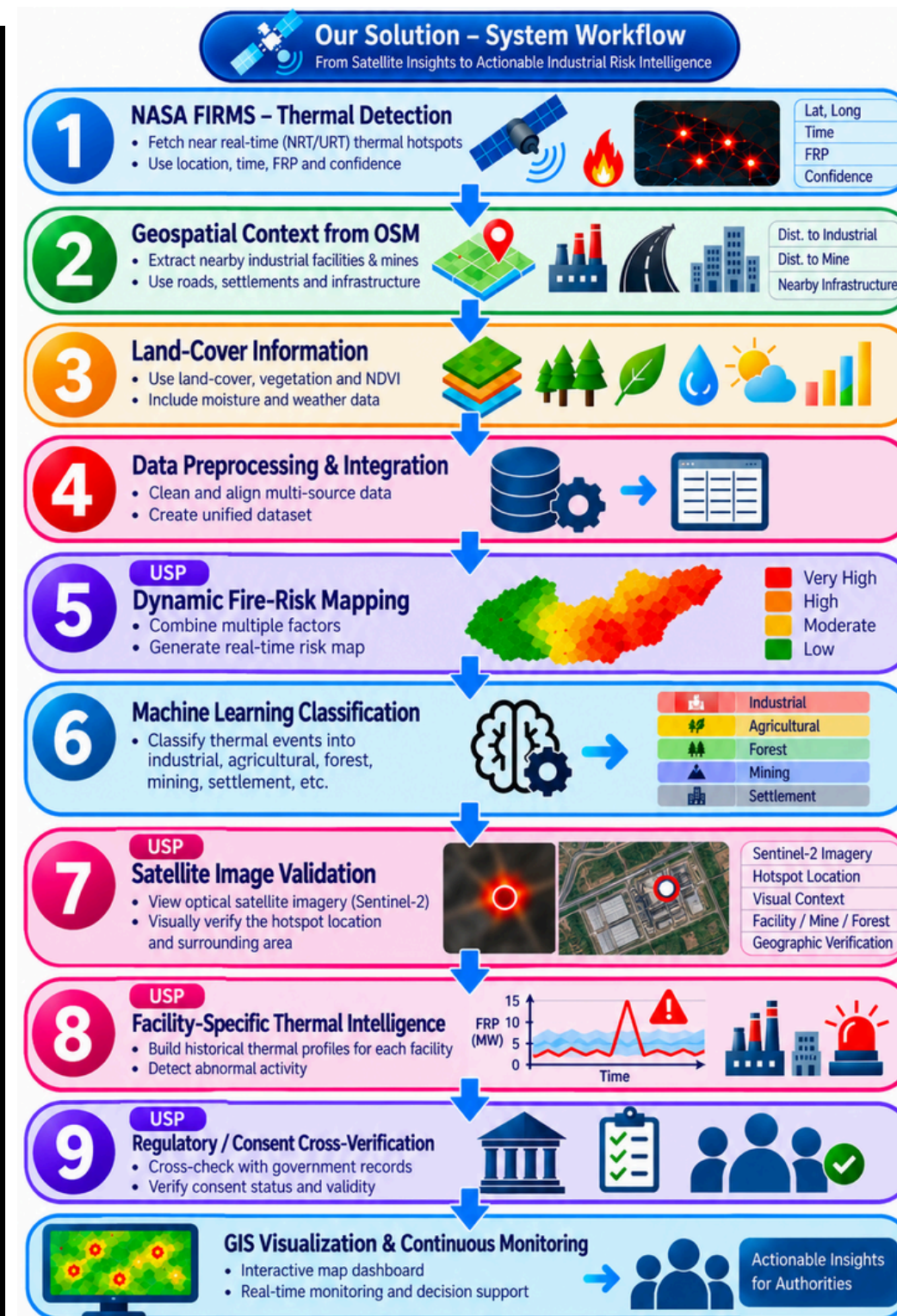


Proposed Idea / Approach

Current satellite fire-monitoring systems such as NASA FIRMS detect thermal anomalies but cannot reliably distinguish industrial fires and persistent thermal sources from forest, agricultural and other fires. Our system addresses this gap by combining thermal, geographic and land-cover intelligence with AI-based classification.

Following is our Idea/Approach:

- **Multi-Source Thermal Dataset:** Integrates NASA FIRMS, OSM and Land-Cover data to build a contextualized dataset of detected thermal anomalies.
- **Geospatial Feature Engineering:** Extracts thermal, spatial, temporal, proximity and land-cover features around each FIRMS detection.
- **AI Fire Classification Engine:** Uses Machine Learning to learn patterns and classify FIRMS-detected anomalies into industrial and non-industrial fire/thermal-source categories.
- **Persistence Analysis:** Identifies recurring and persistent thermal sources using spatiotemporal patterns.
- **Dynamic Monitoring:** Applies the trained model to NASA FIRMS NRT/URT detections.
- **Satellite Validation:** Enables visual verification of selected hotspots using satellite imagery.
- **GIS Intelligence:** Displays classification, confidence, location, FRP and persistence on an interactive map.



Uniqueness

1. Persistence & Recurrence Analysis

Identifies persistent and recurring thermal sources.

2. Satellite Image Validation

Visually verifies hotspot location and surroundings.

3. Facility-Specific Thermal Intelligence

Detects abnormal thermal activity at facilities.

4. Regulatory / Consent Cross-Verification

Cross-checks anomalies with government records.

How it Addresses the Problem

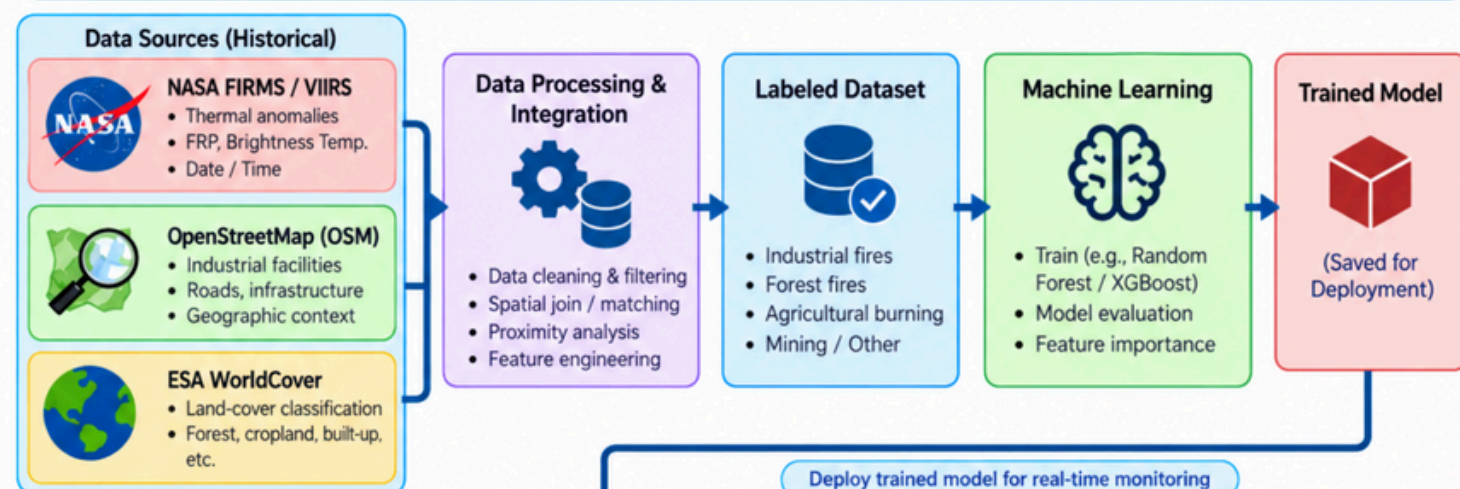
- NASA FIRMS detects initial thermal anomalies.
- OSM + land cover provide facility and environmental context.
- ML classification identifies fire/source categories.
- Dynamic risk mapping highlights areas prone to fire spread.
- Source & origin attribution estimates probable fire origin and source.
- Facility-specific intelligence detects abnormal thermal behaviour.
- Regulatory cross-verification checks government consent and environmental records.
- GIS visualization displays actionable results for authorities.

SYSTEM ARCHITECTURE

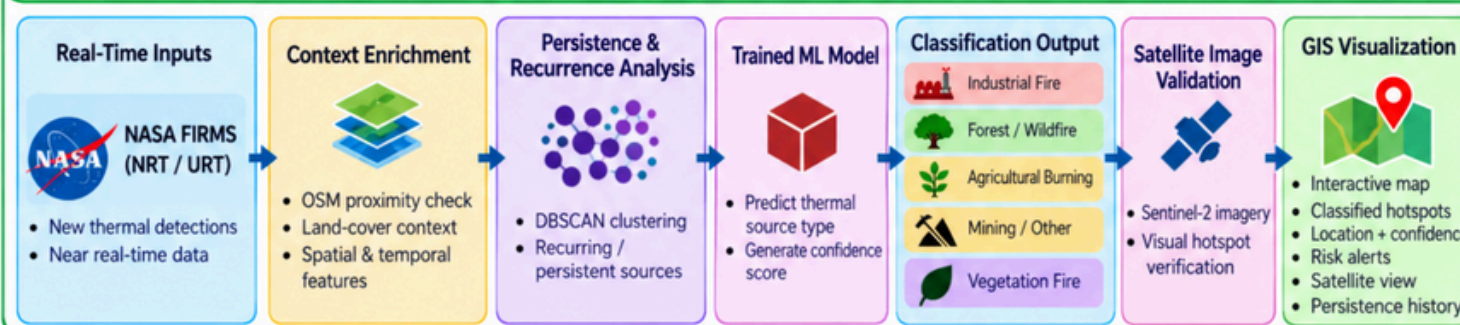
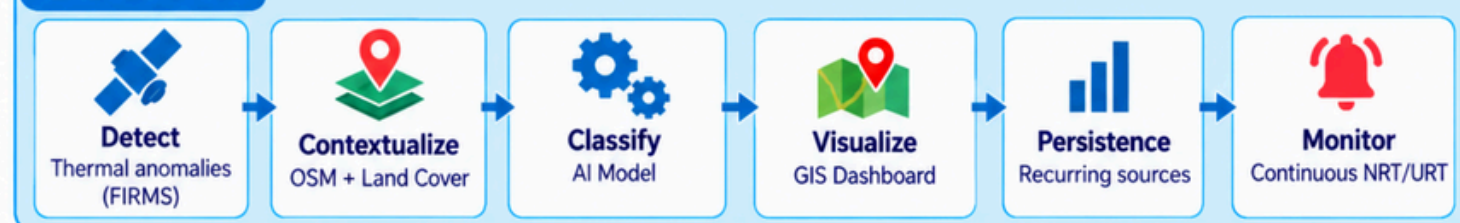
From Raw Thermal Detections to Actionable Intelligence

1. OFFLINE TRAINING (Historical Data)

Build and train the model using historical data (e.g., 2021)

**2. REAL-TIME MONITORING (NRT / URT Data)**

Apply the trained model to new FIRMS detections (e.g., 2022 onwards)

**END OUTPUT****TECHNICAL STACK****Frontend : Web GIS Dashboard**

React + TypeScript | Google Maps API + Leaflet.js | Leaflet-Heat + Chart.js | Vanilla CSS3

Backend : API & ML Inference

FastAPI | Pydantic v2 | GeoJSON

AI/ML Engine : Thermal Classification

Weighted XGBoost | PyTorch | NumPy | Pandas | Feature Engineering

Satellite & Geospatial Data

NASA FIRMS/VIIRS | ESA WorldCover 10m | OpenStreetMap | GeoPandas | Rasterio | Shapely

Persistence Tracking

DBSCAN Spatial Clustering | SciPy cKDTree | Recurrence & Persistence Algorithms



NumPy

XGBoost



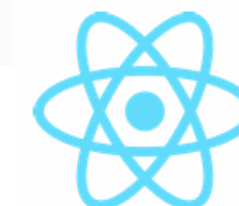
FastAPI



PyTorch



DBSCAN



Feasibility

- **Available Data:** NASA FIRMS, OSM, and WorldCover provide the required input datasets.
- **Technical Capability:** Python, GeoPandas, Rasterio, and Scikit-learn support the complete processing and ML pipeline.
- **No Specialized Hardware:** The system can be developed and tested on standard computing systems.
- **Scalable Processing:** The pipeline can handle increasing thermal detections and larger study areas.

Challenges

- **Incomplete OSM Data:** Missing or outdated facility records can affect classification.
- **Class Imbalance:** Uneven classes can reduce ML performance.
- **Label Uncertainty:** Automated labels may contain noise and require validation.
- **Limited Resolution:** Small or closely located thermal sources may be difficult to distinguish.
- **Evolving Industrial Landscape:** New industries, facility changes and regulatory updates require periodic model updates.

Viability

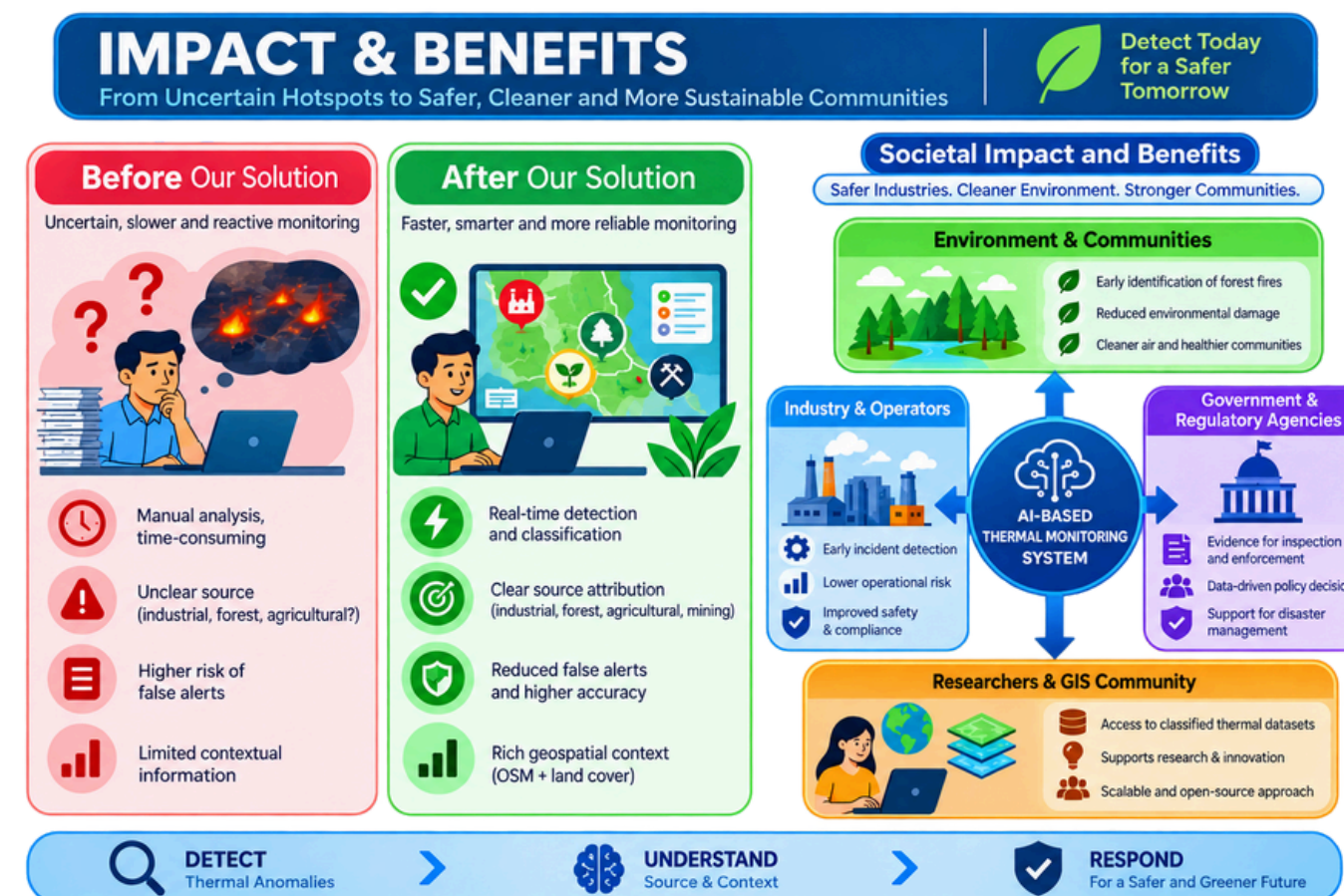
- **Low-Cost Solution:** Uses freely available datasets and open-source technologies, reducing deployment costs.
- **Real-World Use:** Can help identify industrial fires, persistent thermal sources, and potential risks.
- **Continuous Monitoring:** FIRMS NRT/URT data enables regular updates and ongoing monitoring.
- **Future Expansion:** Can be extended with more regions, data sources, and improved ML models.

Strategies to Overcome Challenges

- **Multi-Source Fusion:** Combine FIRMS, OSM and land-cover data.
- **Balanced Training:** Use class balancing and stratified sampling.
- **Label Validation:** Refine automated labels through filtering and manual checks.
- **Feature Fusion:** Combine thermal, spatial, land-cover and temporal features.
- **Model Updating:** Periodically update the model for new industries and regulatory changes.

TARGET AUDIENCE / IMPACT

- ❖ **Industrial Operators:**
Early identification of abnormal thermal activity around industrial facilities.
- ❖ **Disaster Management:**
Faster prioritization of potentially hazardous thermal events for timely response.
- ❖ **Forest & Environmental Agencies:**
Better separation of industrial thermal sources from forest and natural fires.
- ❖ **Regulatory Authorities:**
Geospatial evidence to support inspection, investigation and data-driven decisions.
- ❖ **GIS & Monitoring Teams:**
Unified map-based view of classified and persistent thermal sources for continuous monitoring.

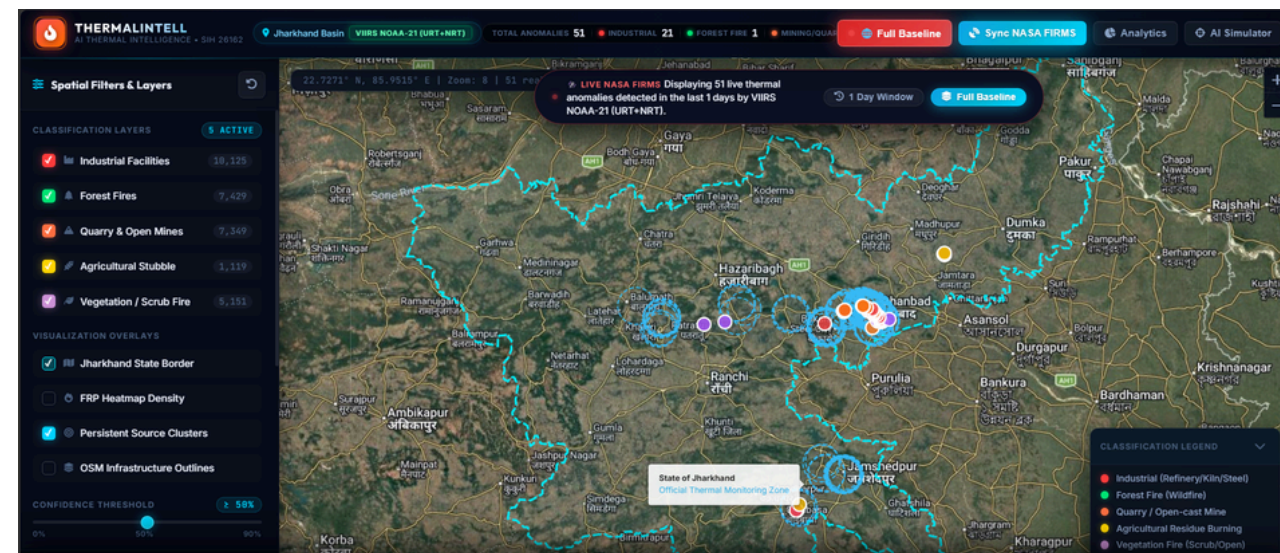
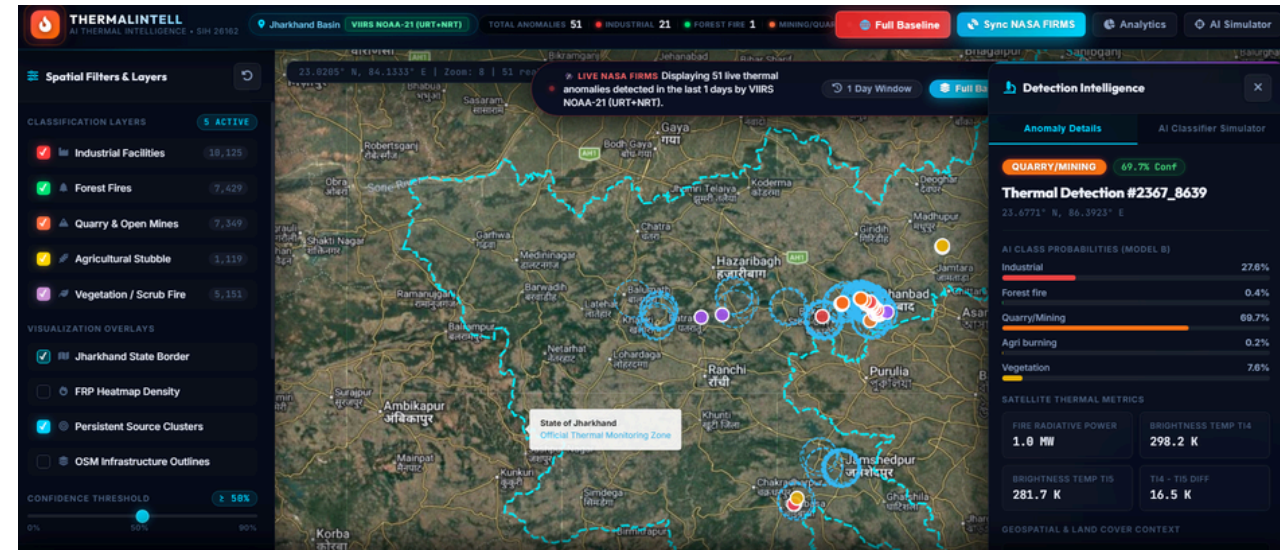


Benefits of the Solution

- 1. Early detection & monitoring:** Automatically identifies industrial fires and persistent thermal hotspots using satellite data, enabling faster response.
- 2. Environmental & public safety:** Helps monitor forest, vegetation, and industrial thermal activity to identify potential environmental and safety risks.
- 3. Cost-effective & scalable:** Reduces dependence on extensive ground monitoring and can be scaled to monitor large regions through FIRMS, ML, and GIS.

Git Hub And Live Demo

<https://github.com/DeClemo/THERMALINTELL>
<https://thermalintell.onrender.com/>



Data Sources

► NASA FIRMS — Fire Information for Resource Management System.

firms.modaps.eosdis.nasa.gov/active_fire/

► Giglio et al. / NASA MODIS & VIIRS Active Fire Research.

doi.org/10.1016/j.rse.2015.11.030

► ESA WorldCover.

esa-worldcover.org/en/data-access

► Copernicus Sentinel-2.

dataspace.copernicus.eu/datacollections/copernicus-sentinel-missions/sentinel-2

► OpenStreetMap.

www.openstreetmap.org/

► Lundberg & Lee, 2017 — A Unified Approach to Interpreting Model Predictions.

proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html